

1.4 Inverses; Algebraic Properties of Matrices

This section studies the algebraic properties of matrix operations: many rules of arithmetic for real numbers carry over to matrices, but some do not.

Properties of Matrix Addition and Scalar Multiplication

Theorem 1.4.1 – Properties of Matrix Arithmetic

Whenever the sizes permit the indicated operations:

- | | |
|---------------------------------|--------------------------------------|
| (a) $A + B = B + A$ | [Commutative law for addition] |
| (b) $A + (B + C) = (A + B) + C$ | [Associative law for addition] |
| (c) $A(BC) = (AB)C$ | [Associative law for multiplication] |
| (d) $A(B + C) = AB + AC$ | [Left distributive law] |
| (e) $(B + C)A = BA + CA$ | [Right distributive law] |
| (f) $A(B - C) = AB - AC$ | |
| (g) $(B - C)A = BA - CA$ | |
| (h) $a(B + C) = aB + aC$ | |
| (i) $a(B - C) = aB - aC$ | |
| (j) $(a + b)C = aC + bC$ | |
| (k) $(a - b)C = aC - bC$ | |
| (l) $a(bC) = (ab)C$ | |
| (m) $a(BC) = (aB)C = B(aC)$ | |

To prove any part, show both sides have the same size and matching entries. As a sample, we prove (d).

Proof (d) Since B, C must have the same size ($m \times n$, say) for $B + C$ to be defined, and A must then have m columns, A has size $r \times m$, making $A(B + C)$ an $r \times n$ matrix – as is $AB + AC$. For entries, with $A = [a_{ij}]$, $B = [b_{ij}]$, $C = [c_{ij}]$:

$$(A(B + C))_{ij} = a_{i1}(b_{1j} + c_{1j}) + \cdots + a_{im}(b_{mj} + c_{mj}) = (AB)_{ij} + (AC)_{ij} = (AB + AC)_{ij} \blacksquare$$

(There are three ways to prove two same-size matrices equal: matching entries, matching row vectors, or matching column vectors.)

Remark Because addition and multiplication are associative, sums and products of three matrices can be written as $A + B + C$ and ABC without parentheses – any placement of parentheses gives the same result.

EXAMPLE 1 | Associativity of Matrix Multiplication

For $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$:

$$AB = \begin{bmatrix} 8 & 5 \\ 20 & 13 \\ 2 & 1 \end{bmatrix}, \quad BC = \begin{bmatrix} 10 & 9 \\ 4 & 3 \end{bmatrix}$$

$$(AB)C = \begin{bmatrix} 18 & 15 \\ 46 & 39 \\ 4 & 3 \end{bmatrix} = A(BC)$$

confirming Theorem 1.4.1(c).

Properties of Matrix Multiplication

Unlike real numbers, $AB = BA$ can fail in three ways: (1) AB defined but BA not; (2) both defined but of different sizes; (3) both defined, same size, but unequal (next example).

EXAMPLE 2 | Order Matters in Matrix Multiplication

For $A = \begin{bmatrix} -1 & 0 \\ 2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$: $AB = \begin{bmatrix} -1 & -2 \\ 11 & 4 \end{bmatrix}$ but $BA = \begin{bmatrix} 3 & 6 \\ -3 & 0 \end{bmatrix}$, so $AB \neq BA$.

Hence **matrix multiplication is not commutative** in general; when $AB = BA$ happens to hold, A and B are said to **commute**.

Zero Matrices

A matrix of all zeros is a **zero matrix**, denoted 0 (or $0_{m \times n}$ if the size matters). For same-size $A, 0$: $A + 0 = 0 + A = A$, paralleling $a + 0 = a$.

Theorem 1.4.2 – Properties of Zero Matrices

For a scalar c and matrices of compatible size:

$$(a) A + 0 = 0 + A = A \quad (b) A - 0 = A \quad (c) A - A = A + (-A) = 0$$

$$(d) 0A = 0 \quad (e) \text{ If } cA = 0, \text{ then } c = 0 \text{ or } A = 0.$$

Two more real-number laws fail for matrices: the **cancellation law** ($ab = ac, a \neq 0 \Rightarrow b = c$) and the **zero-product law** ($ab = 0 \Rightarrow a = 0$ or $b = 0$).

EXAMPLE 3 | Failure of the Cancellation Law

For $A = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 5 \\ 3 & 4 \end{bmatrix}$: $AB = AC = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$, yet $A \neq 0$ and $B \neq C$ – so cancellation fails in general.

EXAMPLE 4 | A Zero Product with Nonzero Factors

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 3 & 7 \\ 0 & 0 \end{bmatrix}: AB = 0 \text{ though } A \neq 0, B \neq 0.$$

Identity Matrices

A square matrix with 1's on the main diagonal and 0's elsewhere is an **identity matrix**, denoted I (or I_n for size n). For any $m \times n$ matrix A :

$$AI_n = A \quad \text{and} \quad I_m A = A$$

paralleling $a \cdot 1 = 1 \cdot a = a$.

Theorem 1.4.3

If R is the reduced row echelon form of an $n \times n$ matrix A , then either R has a row of zeros or $R = I_n$.

Proof Either the last row of R is all zeros, or it isn't – in which case every row has a leading 1, and since these occur progressively farther right down the matrix, each must lie on the main diagonal; the rest of each such column is zero, so $R = I_n$. ■

Inverse of a Matrix

Every nonzero real a has a reciprocal a^{-1} with $a \cdot a^{-1} = a^{-1} \cdot a = 1$. The matrix analog:

Definition 1

If A is square and there is a same-size B with $AB = BA = I$, then A is **invertible (nonsingular)** and B is an **inverse** of A . If no such B exists, A is **singular**.

Since $AB = BA = I$ is symmetric in A, B , if B is an inverse of A then A is an inverse of B – they are **inverses of one another**.

EXAMPLE 5 | An Invertible Matrix

$$\text{For } A = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix}, B = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}: AB = BA = I, \text{ so each is the other's inverse.}$$

EXAMPLE 6 | A Class of Singular Matrices

A square matrix with a zero row or column is singular. E.g., for $A = \begin{bmatrix} 1 & 4 & 0 \\ 2 & 5 & 0 \\ 3 & 6 & 0 \end{bmatrix}$ with columns $\mathbf{c}_1, \mathbf{c}_2, \mathbf{0}$:
for any 3×3 matrix B , $BA = B[\mathbf{c}_1 \ \mathbf{c}_2 \ \mathbf{0}] = [B\mathbf{c}_1 \ B\mathbf{c}_2 \ \mathbf{0}]$ has a zero column, so $BA \neq I$ – A is singular.

Properties of Inverses

Can an invertible matrix have more than one inverse? No:

Theorem 1.4.4

If B and C are both inverses of A , then $B = C$.

Proof $BA = I \Rightarrow (BA)C = IC = C$; but also $(BA)C = B(AC) = BI = B$, so $B = C$. ■

So we may speak of “the” inverse of A , written A^{-1} :

$$AA^{-1} = I \quad \text{and} \quad A^{-1}A = I \quad (1)$$

(**Warning:** A^{-1} is not $1/A$ – division by matrices is not defined.)

Theorem 1.4.5

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is invertible if and only if $ad - bc \neq 0$, in which case

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad (2)$$

The quantity $ad - bc$ is the **determinant** of A , written $\det(A)$ or $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ – the product of the main-diagonal entries minus the product of the off-diagonal entries.

EXAMPLE 7 | Calculating the Inverse of a 2×2 Matrix

(a) $A = \begin{bmatrix} 6 & 1 \\ 5 & 2 \end{bmatrix}$: $\det(A) = 7 \neq 0$, so $A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & -1 \\ -5 & 6 \end{bmatrix} = \begin{bmatrix} \frac{2}{7} & -\frac{1}{7} \\ -\frac{5}{7} & \frac{6}{7} \end{bmatrix}$.

(b) $A = \begin{bmatrix} -1 & 2 \\ 3 & -6 \end{bmatrix}$: $\det(A) = (-1)(-6) - (2)(3) = 0$, so A is not invertible.

EXAMPLE 8 | Solution of a Linear System by Matrix Inversion

To solve $u = ax + by$, $v = cx + dy$ for x, y : write $\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$. If $ad - bc \neq 0$, multiply on the left by the inverse:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \quad \Rightarrow \quad \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

giving $x = \frac{du - bv}{ad - bc}$, $y = \frac{av - cu}{ad - bc}$.

Theorem 1.4.6

If A, B are invertible and the same size, then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.

Proof $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$, and similarly $(B^{-1}A^{-1})(AB) = I$. ■

This extends to any number of factors: *a product of invertible matrices is invertible, and its inverse is the product of the inverses in reverse order.*

EXAMPLE 9 | The Inverse of a Product

For $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}$: $AB = \begin{bmatrix} 7 & 6 \\ 9 & 8 \end{bmatrix}$, $(AB)^{-1} = \begin{bmatrix} 4 & -3 \\ -\frac{9}{2} & \frac{7}{2} \end{bmatrix}$; and $A^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & \frac{3}{2} \end{bmatrix}$, so $B^{-1}A^{-1} = \begin{bmatrix} 4 & -3 \\ -\frac{9}{2} & \frac{7}{2} \end{bmatrix} = (AB)^{-1}$, confirming Theorem 1.4.6.

Powers of a Matrix

For square A : $A^0 = I$ and $A^n = \underbrace{AA \cdots A}_{n \text{ factors}}$; if A is invertible, $A^{-n} = (A^{-1})^n = \underbrace{A^{-1}A^{-1} \cdots A^{-1}}_{n \text{ factors}}$. The usual exponent laws hold: $A^r A^s = A^{r+s}$ and $(A^r)^s = A^{rs}$.

Theorem 1.4.7

If A is invertible and $n \geq 0$ an integer:

- (a) A^{-1} is invertible and $(A^{-1})^{-1} = A$.
- (b) A^n is invertible and $(A^n)^{-1} = A^{-n} = (A^{-1})^n$.
- (c) kA is invertible for nonzero scalar k , and $(kA)^{-1} = k^{-1}A^{-1}$.

Proof (c) By Theorem 1.4.1(m),(l): $(kA)(k^{-1}A^{-1}) = k^{-1}(kA)A^{-1} = (k^{-1}k)AA^{-1} = (1)I = I$, and similarly the other way. ■

EXAMPLE 10 | Properties of Exponents

For $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$, $A^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$: $A^{-3} = (A^{-1})^3 = \begin{bmatrix} 41 & -30 \\ -15 & 11 \end{bmatrix}$, and directly $A^3 = \begin{bmatrix} 11 & 30 \\ 15 & 41 \end{bmatrix}$ gives $(A^3)^{-1} = \begin{bmatrix} 41 & -30 \\ -15 & 11 \end{bmatrix} = (A^{-1})^3$, as Theorem 1.4.7(b) guarantees.

EXAMPLE 11 | The Square of a Matrix Sum

Since real numbers commute, $(a+b)^2 = a^2 + 2ab + b^2$; but for matrices, $(A+B)^2 = A^2 + AB + BA + B^2$ in general, simplifying to $A^2 + 2AB + B^2$ only when A, B commute.

Matrix Polynomials

For square A ($n \times n$) and polynomial $p(x) = a_0 + a_1x + \cdots + a_mx^m$, define the **matrix polynomial**

$$p(A) = a_0I + a_1A + a_2A^2 + \cdots + a_mA^m \quad (3)$$

(substituting A for x and a_0I for the constant term).

EXAMPLE 12 | A Matrix Polynomial

For $p(x) = x^2 - 2x - 5$, $A = \begin{bmatrix} -1 & 2 \\ 1 & 3 \end{bmatrix}$:

$$p(A) = A^2 - 2A - 5I = \begin{bmatrix} 3 & 4 \\ 2 & 11 \end{bmatrix} - \begin{bmatrix} -2 & 4 \\ 2 & 6 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Remark Since $A^r A^s = A^{s+r} = A^s A^r$, powers of A commute, so any two matrix polynomials in the same A commute:

$$p_1(A)p_2(A) = p_2(A)p_1(A) \quad (4)$$

Properties of the Transpose

Theorem 1.4.8

Whenever the sizes permit:

$$\begin{aligned} \text{(a)} \quad (A^T)^T &= A & \text{(b)} \quad (A + B)^T &= A^T + B^T & \text{(c)} \quad (A - B)^T &= A^T - B^T \\ \text{(d)} \quad (kA)^T &= kA^T & \text{(e)} \quad (AB)^T &= B^T A^T \end{aligned}$$

Parts (a)–(d) follow directly from “transposing swaps rows and columns.” Part (e) extends to any number of factors: *the transpose of a product is the product of the transposes in reverse order.*

Theorem 1.4.9

If A is invertible, so is A^T , and $(A^T)^{-1} = (A^{-1})^T$.

Proof By Theorem 1.4.8(e) and $I^T = I$: $A^T(A^{-1})^T = (A^{-1}A)^T = I^T = I$, and $(A^{-1})^T A^T = (AA^{-1})^T = I^T = I$. ■

EXAMPLE 13 | Inverse of a Transpose

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ invertible ($\det A = ad - bc \neq 0$), $A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$ has the same determinant, so it too is invertible, and by Theorem 1.4.5, $(A^T)^{-1}$ works out to be exactly $(A^{-1})^T$ – confirming Theorem 1.4.9.

Exercise Set 1.4

- For $A = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix}$, $C = \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}$, $a = 4, b = -7$, verify:
 - the associative law for addition
 - the associative law for multiplication
 - the left distributive law
 - $(a + b)C = aC + bC$

- Using the matrices from Exercise 1, verify:

$$\text{(a)} \quad (A^T)^T = A \quad \text{(b)} \quad (AB)^T = B^T A^T$$

- Use Theorem 1.4.5 to find the inverse of $A = \begin{bmatrix} 2 & -3 \\ 4 & 4 \end{bmatrix}$.

- Find the inverse of $\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$.

- For $A = \begin{bmatrix} 2 & -3 \\ 4 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, verify $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$.

- Given $(7A)^{-1} = \begin{bmatrix} -3 & 7 \\ 1 & -2 \end{bmatrix}$, find A .

19. For $A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$, compute (a) A^3 (b) A^{-3} (c) $A^2 - 2A + I$.
21. For $A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$, compute $p(A)$ for
 (a) $p(x) = x - 2$ (b) $p(x) = 2x^2 - x + 1$ (c) $p(x) = x^3 - 2x + 1$
23. For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, find all a, b, c, d for which A, B commute.
25. Use the method of Example 8 to find the unique solution of
 $3x_1 - 2x_2 = -1$, $4x_1 + 5x_2 = 3$
30. Verify $p(A) = p_1(A)p_2(A)$ for $p(x) = x^2 - 9 = (x + 3)(x - 3)$ and an arbitrary square matrix A .
31. (a) Give an example of 2×2 matrices with $(A + B)(A - B) \neq A^2 - B^2$.
 (b) State a valid formula for $(A + B)(A - B)$.
 (c) What condition on A, B allows $(A + B)(A - B) = A^2 - B^2$?
32. The equation $a^2 = 1$ has exactly two real solutions. Find at least eight solutions of $A^2 = I_3$. [*Hint*: look for solutions with all off-diagonal entries zero.]
33. (a) Show that if $A^2 + 2A + I = 0$, then A is invertible; find A^{-1} .
 (b) Show that if $p(x)$ has nonzero constant term and $p(A) = 0$, then A is invertible.
34. Is it possible for A^3 to equal the identity matrix without A being invertible? Explain.
35. Can a matrix with a zero row or column have an inverse? Explain.
36. Can a matrix with two identical rows or columns have an inverse? Explain.
37. Determine whether $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ is invertible, and if so find A^{-1} . [*Hint*: solve $AX = I$ for X by equating corresponding entries.]
39. Simplify, assuming A, B, C, D invertible: $(AB)^{-1}(AC^{-1})(D^{-1}C^{-1})^{-1}D^{-1}$.
41. Show that if R is $1 \times n$ and C is $n \times 1$, then $RC = \text{tr}(CR)$.
43. (a) Show that if A is invertible and $AB = AC$, then $B = C$.
 (b) Explain why part (a) and Example 3 do not contradict each other.
45. (a) Show that if $A, B, A + B$ are invertible matrices of the same size, then $A(A^{-1} + B^{-1})B(A + B)^{-1} = I$.
 (b) What does part (a) tell you about the matrix $A^{-1} + B^{-1}$?
46. A square matrix A is **idempotent** if $A^2 = A$.
 (a) Show that if A is idempotent, so is $I - A$.
 (b) Show that if A is idempotent, then $2A - I$ is invertible and is its own inverse.
47. Show that if $A^k = 0$ for some positive integer k , then $I - A$ is invertible and $(I - A)^{-1} = I + A + A^2 + \cdots + A^{k-1}$.

48. Show that $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ satisfies $A^2 - (a + d)A + (ad - bc)I = 0$.

51. **(Working with Proofs)** Prove Theorem 1.4.1(a).